

STAGE 1

Graph Ingestion & Attacks

User-Item Bipartite Graph

$$\mathcal{G} = (\mathcal{U}, \mathcal{I}, \mathcal{E}), \tilde{R} \in \{0, 5\}^{|\mathcal{U}| \times |\mathcal{I}|}$$

- Clean Interactions: Solid Edges
- Injected Shilling: Bandwagon (15%)
- Temporal Review-Bombing Bursts

$$\mathbf{A} \in \mathbb{R}^{|\mathcal{U}| \times |\mathcal{I}|}, \hat{\mathbf{A}} = \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$$

1. MovieLens-100K
(943 users, 1,682 items)
2. Amazon Electronics
(5-Core Sparse Graph)
3. Yelp Academic Dataset
(Dense Burst Reviews)

K-Core Graph Pruning ($k \geq 5$)
Train / Val / Test (80/10/10 Split)
Pareto Head/Tail Indexing

$$\mathcal{I}_{\text{head}} \cup \mathcal{I}_{\text{tail}} = \mathcal{I}$$

STAGE 2

MIRF Reliability Filter

View 1: RRFN Noise Matrix T

$$T_{\lambda}^{\dagger} = (T^T T + \lambda I)^{-1} T^T$$

$$R_{\text{RRFN}} = \exp\left(-\frac{\| \tilde{r} - \mathbb{E}[Y^*] \|^2}{2\sigma^2}\right)$$

View 2: LLM Semantic Audit

Async Prompt Sentiment Verification
Thread-Safe SQLite WAL Persistent Cache

$$R_{\text{LLM}}(u, i) \in [0, 1] \text{ (}\mathcal{O}(1)\text{ Lookup)}$$

View 3: 2D BLAS Burst Detector

$$R_{\text{bomb}} = 1 - \sigma(\omega_1 Z_{\text{temp}} + \omega_2 S_{\text{pol}})$$

Vectorized Matrix $W @ X$ (<25ms)

$$w_{ui} = \alpha R_{\text{RRFN}} + \beta R_{\text{LLM}} + \gamma R_{\text{bomb}}$$

Continuous Sample Reliability $w_{ui} \in [0, 1]$

STAGE 3

Risk-Aware GCL & Models

Dual Contrastive Graph Views

- SimGCL: $\mathbf{z}' = \mathbf{e} + \varepsilon \cdot \text{sgn}(\mathbf{e}) \odot \Delta'$
- SGL: Structural Dropout $\hat{\mathbf{A}}' = (\mathbf{M}' \odot \mathbf{A}) \mathbf{D}'^{-1}$

$$\mathcal{L}_{\text{cl}} = -\sum w_i \log \frac{\exp(\text{sim}(\mathbf{z}', \mathbf{z}'')/\tau)}{\sum \exp(\cdot)}$$

Dual Partition Optimization

Head Model $\mathcal{M}_{\text{pop}}$ (Trained on $\mathcal{I}_{\text{head}}$)

$$\mathcal{L}_{\text{total}}^{\text{head}} = \mathcal{L}_{\text{risk}}^{\text{head}} + \lambda_{\text{cl}} \mathcal{L}_{\text{cl}}^{\text{head}}$$

Tail Model $\mathcal{M}_{\text{tail}}$ (Trained on $\mathcal{I}_{\text{tail}}$)

$$\mathcal{L}_{\text{total}}^{\text{tail}} = \mathcal{L}_{\text{risk}}^{\text{tail}} + \lambda_{\text{cl}} \mathcal{L}_{\text{cl}}^{\text{tail}}$$

Backbones: NeuMF | LightGCN | VaeCF
SimGCL (Latent) | SGL (Structural)

$$\mathbf{s}_{\text{head}} \in \mathbb{R}^{|\mathcal{I}_{\text{head}}|}, \quad \mathbf{s}_{\text{tail}} \in \mathbb{R}^{|\mathcal{I}_{\text{tail}}|}$$

STAGE 4

SUPER Calibrated Merging

Denoised Blueprint B_U

$$B_U = \{i \mid w_{ui} \geq \theta_{\text{trust}}\}$$

Purges Injected Shilling Edges
Zero- k Guarded History

Tail Inclination α_U

$$\alpha_U = \frac{\sum_{i \in B_U \cap \mathcal{I}_{\text{tail}}} w_{ui}}{|B_U| + \varepsilon}$$

Personalized Tail Quota Allocation
Robust to Head Bandwagon Attacks

Calibrated Top-K List

$$K_{\text{tail}} = \lfloor \alpha_U K \rfloor, \quad K_{\text{head}} = K - K_{\text{tail}}$$

Merge $C_{\text{head}} \setminus B_U$ & $C_{\text{tail}} \setminus B_U$
Seen-Item Deduplication

$$\mathcal{R}_U = [i_1, i_2, \dots, i_K]$$

STAGE 5: Comprehensive Empirical Validation & Financial Simulation

Academic Metrics:
Recall@10: 0.2067
nDCG@10: 0.1014

Adversarial Defense:
ROC-AUC: 0.7265 (+45%)
PR-AUC: 0.6840

Catalog Calibration:
LTC@10: 0.1590
Novelty: 11.94

A/B Business Simulation:
CTR Lift: +3.64% ($p < 0.001$)
GMV Lift: +\$428,500 / yr

Inference Speed:
2D BLAS Scorer: <25ms
Acceleration: >175x